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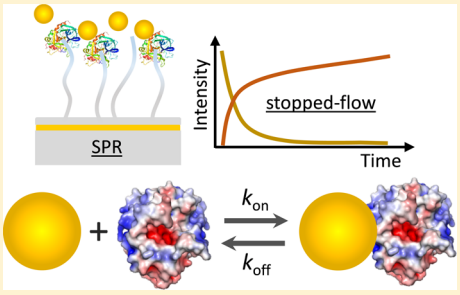
Mechanistic Insights into Ultrasmall Gold Nanoparticle−Protein Interactions through Measurement of Binding Kinetics

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ABSTRACT: Gold nanoparticles (NPs) in the ultrasmall size regime provide a new paradigm in the way that nanomaterials can be used to regulate protein structure and function. However, the rational design of ultrasmall NPs as viable synthetic effectors of protein function requires detailed quantitative understanding of their biomolecular interactions. Herein, we focused on the kinetics of NP−protein complexation—an often neglected factor in studies at the bio-nanointerface—to gain molecular-level insights into mechanisms of interactions. The protein α-thrombin and ultrasmall gold NPs coated with *p*-mercaptobenzoic acid (AuMBA) and glutathione (AuGSH) were used as model systems in our studies. Binding kinetics was quantified by surface plasmon resonance (SPR) biosensing and stopped-flow spectroscopy. The results revealed strong and weak interactions of AuMBA and AuGSH toward thrombin (K_D ∼30 nM and 20 μM, respectively), as well as "fast" and "slow" association kinetics (k_{on} ∼10^6−10^7 and 10^4 M^{-1} s^{-1}, respectively). The significantly smaller k_{on} for AuGSH implied the presence of a larger energy barrier along the association pathway, presumably related to the penalty required to remove interfacial ions. Analysis of dissociation reactions revealed that thrombin interactions with both NPs formed transient, weakly adhesive complexes characterized by short residence times ($t_r = 1/k_{off}$ ∼0.1−16 s). Interestingly, the reverse reactions were best described by multiple dissociation processes suggesting a heterogeneous population of complexes stabilized to different extent.



INTRODUCTION

Synthetic nanosized particles (NPs) interacting with biomolecules, cells, and tissues provide a new paradigm for the control of biological functions.[1−3] Recent experimental and theoretical works have indicated that gold NPs in the ultrasmall size regime (<3 nm diameter) may constitute particularly attractive model systems to regulate the structure and function of proteins and other biological entities through controlled interactions.[4−8] Ultrasmall gold NPs can be prepared with well-defined sizes and excellent degrees of uniformity, while their surface chemistries can be modified almost at will to impart them with distinct physicochemical characteristics and desired functionalities.[9−11] Given the proper choice of surface chemistry, ultrasmall NPs can selectively target unique regions on a protein surface since they are smaller than most proteins.[4,6,12] Moreover, unlike the adsorption of proteins onto much larger NPs,[13−18] the interactions of proteins with similar-sized ultrasmall NPs more closely resemble a typical biomolecular complexation event.[4,6,19,20]

In order to fully realize the potential of ultrasmall gold NPs as viable effectors of protein structure and function, it is essential to quantitatively understand their molecular interactions with proteins. Usually, characterization of NP−protein complexation entails determination of the equilibrium dissociation constant, K_D.[21−24] Although K_D is undoubtedly an important thermodynamic quantity describing complex formation between two interacting partners, equilibrium properties alone bring limited understanding about molecular mechanisms of interactions. On the other hand, knowledge of the individual association (k_{on}) and dissociation (k_{off}) rate constants underlying K_D ($K_D = k_{off}/k_{on}$) can provide unique insights into the interaction mechanisms.[20,25−28] k_{on}, in particular, carries information about the molecular events along the association pathway toward the final bound complex, whereas k_{off} is inversely proportional to complex residence time ($t_r = 1/k_{off}$) and reports on the stability of the binding interface.

Recently, we have characterized for the first time the binding kinetics of ultrasmall gold NPs with proteins.[20] The protein CrataBL and ultrasmall gold NPs passivated with *p*-mercaptobenzoic acid (AuMBA) and glutathione (AuGSH) were used as model systems in our studies. Characterization by surface plasmon resonance (SPR) revealed strong and weak binding affinities of AuMBA and AuGSH toward CrataBL (K_D ∼30 nM and 30 μM, respectively), as well as "fast" and "slow" association kinetics (k_{on} ∼10^6 and 10^4 M^{-1} s^{-1}, respectively). The smaller k_{on} for AuGSH suggested the presence of a larger energy barrier along the association pathway slowing down the interactions. Interestingly, computer simulations showed that

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this barrier originated from the greater structural and chemical complexity of the AuGSH passivating layer, with the more complex ionic atmosphere surrounding AuGSH yielding a higher energy penalty for desolvation.

Here, we report a new set of kinetic measurements to further advance our mechanistic understanding of NP−protein complexation in the ultrasmall size regime. Toward that end, ultrasmall AuMBA and AuGSH were used here as prototype NPs displaying "strong" and "weak" interactions with human α-thrombin. This protein represents an attractive model system to perform binding studies with NPs, as it contains two well-defined and highly positively charged exosite domains situated on opposite ends of the molecule[29] where negatively charged NPs can bind. In addition, it can be easily functionalized at its active-site region with either biotin or fluorescence probes.[30,31] Measurements of binding kinetics were performed in this work by surface plasmon resonance (SPR) using biotin-labeled thrombin immobilized onto a streptavidin-coated surface. For the first time for ultrasmall gold particles, we extend the kinetic studies to stopped-flow spectroscopy in solution using fluorescein-labeled thrombin.

EXPERIMENTAL SECTION

Materials and Reagents. Human α-thrombin and fluorescein−PPACK (FITC−PPACK) were purchased from Molecular Innovations (Novi, MI). Biotinylated thrombin was obtained from Haematologic Technologies (Essex Junction, VT). The aptamers HD1 and HD22 were synthesized by Exxtend (Campinas, SP). Sensor chip SA with preimmobilized streptavidin was purchased from GE Healthcare Life Sciences (Piscataway, NJ). The remaining reagents were obtained from Sigma-Aldrich (Milwaukee, WI). Thrombin was labeled stoichiometrically at its active site with FITC−PPACK for subsequent use in fluorescence experiments.[30] Labeling was accomplished by mixing a 10-fold molar excess of FITC−PPACK with thrombin for 30 min followed by purification using a Sephadex-G75 gel filtration column. The AuMBA nanoparticles were prepared and characterized as described in detail in several previous reports.[32−35]

Fluorescence Spectroscopy. Fluorescence measurements were performed on a Shimadzu spectrofluorimeter model RF-6000. FITC-labeled thrombin was loaded in a quartz cuvette at a concentration of 100 nM and titrated with AuMBA or AuGSH. The assays were performed at 25 °C in phosphate buffer solution (20 mM, pH 7.2) supplemented with NaCl (0, 50, 150, 300, and 500 mM). Fluorescence spectra were acquired using an excitation wavelength of 495 nm and 3 nm bandpass. The inner-filter effect from the NPs was accounted for by titrating a solution of FITC molecules with identical concentrations of NPs. Corrected quenching curves for FITC-labeled thrombin were then generated by dividing the uncorrected data by the FITC reference curve.[36,37] Titrations in the presence of HD1 and HD22 were carried out in a similar way, except that thrombin was equilibrated with a high concentration of the aptamers (15 and 5 μM, respectively) prior to adding the NPs.

Surface Plasmon Resonance (SPR). SPR measurements were carried out on a Biacore T-200 instrument (GE Healthcare Life Sciences). Biotinylated thrombin (where the biotin label was attached to the active site via the irreversible inhibitor PPACK) was immobilized to a commercial carboxymethylated dextran matrix containing streptavidin. Thrombin immobilization was performed at three different surface densities: 120, 270, and 515 response unit (RU). Experiments were performed at 25 °C using phosphate buffer solution (20 mM, pH 7.2) supplemented with 150 mM NaCl as the running buffer. The NPs were injected in the flow in the concentration range from 1 nM to 20 μM and at a flow rate of 90 μL min^{-1}. The association and dissociation phases were recorded for 200 and 700 s, respectively. Regeneration of the sensor surface between injections was achieved by washing with 0.005% sodium dodecyl sulfate in water followed by a 2 M solution of NaCl in water (wash time = 30 s; flow rate = 30 μL min^{-1}). Correction of bulk refractive index changes was accomplished by subtracting the responses from a reference surface from the raw SPR traces. Data analysis was carried out with the software EVILFIT assuming a continuous surface-site distribution model.[38−40]

Stopped-Flow Spectroscopy. Stopped-flow measurements were performed on an SX20 model instrument (Applied Photophysics, Leatherhead, U.K.) equipped with a 490 nm LED light source, a 500 nm short-pass filter placed at the excitation path, and a 515 nm long-pass filter in the emission path. Samples were prepared in phosphate buffer solution (20 mM) supplemented with 150 mM NaCl. The kinetics of AuMBA−protein association was followed under pseudo-first-order conditions with an FITC−thrombin concentration of 30 nM and nominal AuMBA concentrations in the range from 0.1 to 1.5 μM (after mixing). For each condition, about 12 traces were recorded and fit individually to a single exponential function of the form: $F = A\cdot\exp(-k_{obs}t) + F_\infty$, where F is the measured fluorescence signal, A is the pre-exponential factor, F_∞ is the final fluorescence, and k_{obs} is the observed first-order rate constant. The apparent association rate constant (k_{on}) was calculated from the slope of the line in a plot of k_{obs} versus [AuMBA]. The kinetics of NP−protein dissociation was evaluated by means of a competitive displacement assay wherein preformed complexes of FITC−thrombin with NPs (50 nM and 1 μM, respectively) were irreversibly dissociated by competitive displacement with excess unlabeled thrombin (40 μM). The obtained time courses were modeled by a triple exponential function whose corresponding observed first-order rate constants equaled the apparent dissociation rate constants k_{off1}, k_{off2}, and k_{off3}.

RESULTS AND DISCUSSION

Nanoparticles and α-Thrombin. The synthesis and characterization of AuMBA and AuGSH have been described in a number of previous publications.[20,32−35] Briefly, the NPs have a core diameter of around 2 nm and are highly uniform (standard deviations of size measurements are typically 0.1−0.2 nm as determined by electron microscopy). The NPs are negatively charged and display similar values of ζ potential around −22 ± 0.5 and −22.5 ± 0.7 mV for AuMBA and AuGSH, respectively. The model protein used in this study, human α-thrombin, has a molecular mass of 37 kDa and a roughly spherical shape 5 nm in diameter.[41] The surface charge distribution of thrombin is highly anisotropic as shown by electrostatic surface potential calculations[29,42] (Figure 1). The active-site cleft of thrombin is surrounded by an acidic surface patch and flanked by two highly positively charged domains known as exosites 1 and 2.[29] Importantly, the active site of thrombin can be derivatized with different functionalities via the irreversible inhibitor D-Phe-Pro-Arg-CH_2Cl (FPRCK or PPACK).[30] In this work, thrombin's active site was labeled with FITC−PPACK to facilitate binding studies by fluo-

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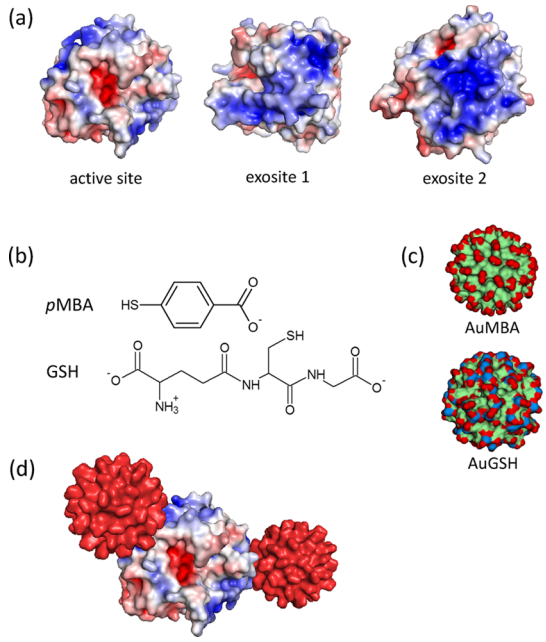


Figure 1. Thrombin and gold NPs. (a) Surface electrostatic potential of α-thrombin (1PPB) scaled from −5 to +5 kT/e (red to blue; calculated with APBS[42]). Thrombin contains two positively charged domains (exosites) flanking the active site. (b) Structure of *p*-mercaptobenzoic acid (*p*MBA) and glutathione (GSH) passivating ligands. (c) Molecular surface representations of the NPs (red: $-CO_2^-$ groups; blue: $-NH_3^+$). The NPs have identical core sizes and similar overall diameters but differ in surface chemistry, charge distribution, and topography. (d) Schematics of NP−thrombin complexation. The negatively charged NPs bind preferentially to the positively charged exosites. See text for details.

rescence quenching titration. Commercially available thrombin derivatized with biotin−PPACK was also used for immobilization onto the SPR sensor surface.

NP−Thrombin Interactions by Fluorescence Quenching Titration. We first carried out binding experiments of AuMBA− and AuGSH−thrombin interactions by fluorescence quenching titration using FITC-labeled thrombin. Because the FITC probe is positioned in the active site, it does not interfere with NP complexation to the exosites (vide infra). We note that the interactions between AuMBA and thrombin have been previously investigated by us in detail.[4] Here, we present new data for AuMBA obtained under identical experimental conditions to AuGSH, thus allowing for a more accurate comparison between the two NP types.

The fluorescence signal of 100 nM thrombin in 150 mM NaCl was completely quenched by the addition of ∼100 nM AuMBA, whereas a 10-fold higher concentration of ∼1 μM was required to approach the same degree of quenching with AuGSH (Figure 2, black squares). This demonstrates much stronger binding of AuMBA to thrombin. Quantification of the fluorescence data was precluded by the multivalent nature of the interactions;[36,37] nevertheless, the stronger binding of AuMBA toward thrombin is in qualitative agreement with the SPR results.

Fluorescence quenching was strongly dependent on the salt concentration, suggesting electrostatically controlled complexation (Figure 3). FITC fluorescence was equally quenched by both AuMBA and AuGSH at [NaCl] = 0−50 mM, whereas at [NaCl] > 150 mM efficient quenching was achieved with AuMBA only. This suggests that NP−thrombin complexation at low salt concentrations is primarily driven by long-range electrostatic forces, whereas at higher salt concentrations the interactions may also be modulated by the local chemical and structural characteristics of the NP and protein surfaces.[20,43,44]

Ultrasmall NPs are smaller than most proteins, thus raising the prospect of selectively targeting defined regions on a protein surface.[4,6,12,45] Previously, we showed that negatively charged AuMBA interacted preferentially around the positively charged exosite domains of thrombin. This was demonstrated with the help of the exosite-directed aptamers HD1 and HD22 in competition experiments with AuMBA.[46,47] The new results obtained for AuMBA were similar to those previously reported by us (Figure 2).[4] Specifically, FITC fluorescence was almost completely quenched in the presence of a single aptamer type, whereas the signal dropped by only ∼15% of its original value at a AuMBA concentration of 1 μM when both aptamers were included simultaneously. This confirmed exosites 1 and 2 as the main binding sites for AuMBA on the protein surface. Similar to AuMBA, AuGSH was also found to selectively bind to the exosites (Figure 2). Interestingly, however, no binding of AuGSH toward exosite 1 could be detected by fluorescence quenching in the presence of the competing exosite 2 ligand, HD22.

To conclude this part, we have established that AuMBA and AuGSH selectively target exosites 1 and 2 on the surface of thrombin. Binding is electrostatically driven while possibly being modulated by additional chemical and structural features of the NP and protein surfaces, especially at higher salt

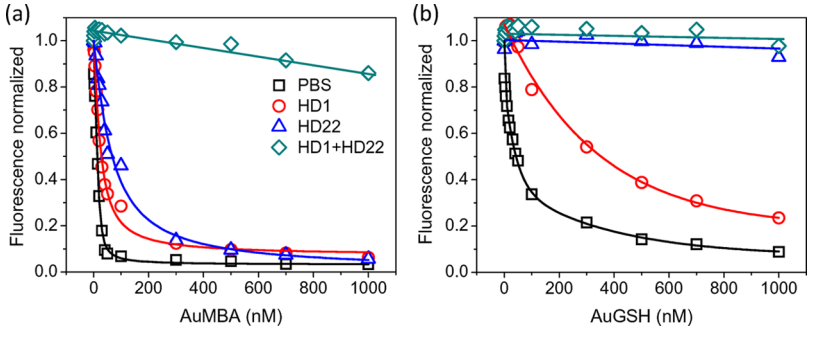


Figure 2. Fluorescence quenching titration of α-thrombin with (a) AuMBA and (b) AuGSH nanoparticles. Data was recorded in phosphate buffer solution supplemented with 150 mM NaCl in the absence (black squares) or presence of the exosite ligands HD1, HD22, and HD1 + HD22. Thrombin, HD1, and HD22 were used at concentrations of 0.1, 15, and 5 μM, respectively. Lines are guides to the eye.

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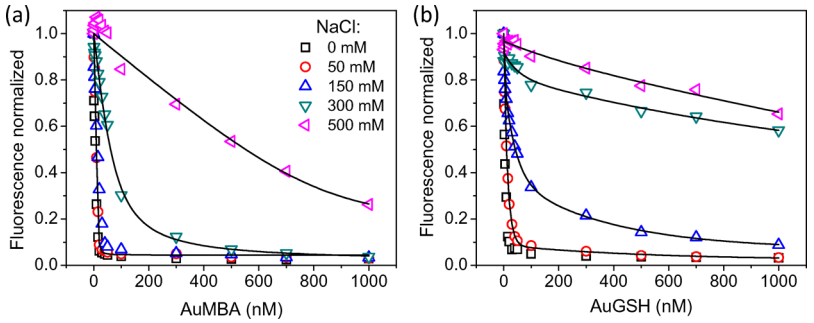


Figure 3. Fluorescence quenching titration of α-thrombin with (a) AuMBA and (b) AuGSH nanoparticles. Data was recorded in phosphate buffer solution supplemented with different NaCl concentrations (0, 50, 150, 300, and 500 mM; see figure legend). Thrombin was used at a concentration of 0.1 μM. Lines are guides to the eye.

concentrations. All things considered, thrombin constitutes an attractive model system to investigate protein interactions with nanomaterials.

Binding Kinetics Probed by Surface Plasmon Resonance. Thrombin was immobilized onto a commercial streptavidin sensor surface through a biotin−PPACK linker attached to its active site. This immobilization strategy keeps thrombin uniformly oriented on the sensor surface with both exosites presumably exposed, in contrast to the random orientation of immobilized proteins obtained with the standard amine coupling method. SPR measurements were performed by flowing AuMBA or AuGSH over the immobilized thrombin using phosphate buffer solution supplemented with 150 mM NaCl as the running buffer (Figure 4). The consistency of the results was verified by the use of different sensor surfaces containing increasing levels of protein immobilization (120, 270, and 515 RU).

The characterization of NP−protein interactions by SPR biosensing can be expected to deviate from the simplest mechanism of a single-site binding. Multivalent binding, nonuniformity of the NP size and surface chemistry, binding avidity, and microheterogeneity of the surface sites are some of the possible factors leading to a multiphasic binding response.[38] Data analysis was therefore performed with a continuous surface-site distribution model, which represents the surface sites as a two-dimensional distribution of affinity and kinetic rate constants.[38−40] This approach has been widely used for the study of protein−protein interactions where a single-site model often does not adequately describe the data.

The binding traces and calculated affinity and rate constant distributions for AuMBA−thrombin interactions are displayed in Figure 5. Results obtained from three different levels of surface immobilization are shown. Integration of the major peak in the distributions (circled areas) yielded the binding parameters listed in Table 1. It can be seen that the AuMBA−thrombin interactions were characterized by an equilibrium dissociation constant in the low nM range (K_D ~31 nM). The calculated association rate constant of ~2 × 10^6 M^{-1} s^{-1} was close to the maximum rate accessible by the SPR technology. The dissociation rate constant of 0.06 s^{-1} implied a short residence time of the final NP−protein complex ($t_r = 1/k_{off}$ ~16 s).

The binding traces and calculated affinity and rate constant distributions for AuGSH−thrombin interactions are displayed in Figure 6. Integration of the major peak in the distributions (circled areas) yielded the binding parameters listed in Table 1. Compared to AuMBA, AuGSH interactions with thrombin were much weaker with an equilibrium dissociation constant in the low μM range (K_D ~19 μM). The association rate constant of ~3 × 10^4 M^{-1} s^{-1} was around 100-fold smaller than that of AuMBA, whereas the dissociation rate constant of ~0.5 s^{-1} was approximately 10-fold higher.

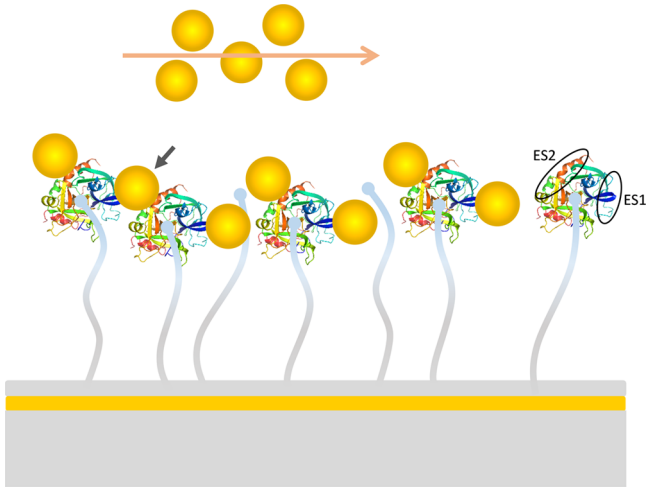


Figure 4. Surface plasmon resonance measurements. Biotinylated thrombin was immobilized to a commercial carboxymethylated dextran matrix preimmobilized with streptavidin (streptavidin molecules not shown for clarity). The biotin label was conjugated to the inhibitor PPACK and attached to thrombin via the active site. This immobilization strategy keeps thrombin presumably uniformly oriented on the sensor surface with both its exosites (ES1 and ES2) well exposed. Nevertheless, heterogeneity of surface binding sites can still occur due to a number of factors including steric hindrance, chemical and physical nonuniformity of the matrix, and clustering of proteins on the surface; the latter can lead to avidity effects through the binding of a single NP to two proteins simultaneously (see arrow mark). The experiments were performed at increasing levels of immobilization density: 120, 270, and 515 RU. The NPs were injected in the flow in the concentration range from 1 nM to 20 μM. Phosphate buffer supplemented with 150 mM NaCl was used as the running buffer. Further details are given in the Experimental Section.

It should be noted that the absolute values of the affinity and kinetic constants determined by SPR can be affected by mass transport limitations or even by the surface itself, which might be viewed as a third component modulating complex formation. Nonetheless, the calculated affinity and rate constant distributions for both AuMBA and AuGSH were remarkably similar as a function of thrombin immobilization density (Figures 5 and 6), which therefore supports the reliability of our measurements, especially in what concerns the

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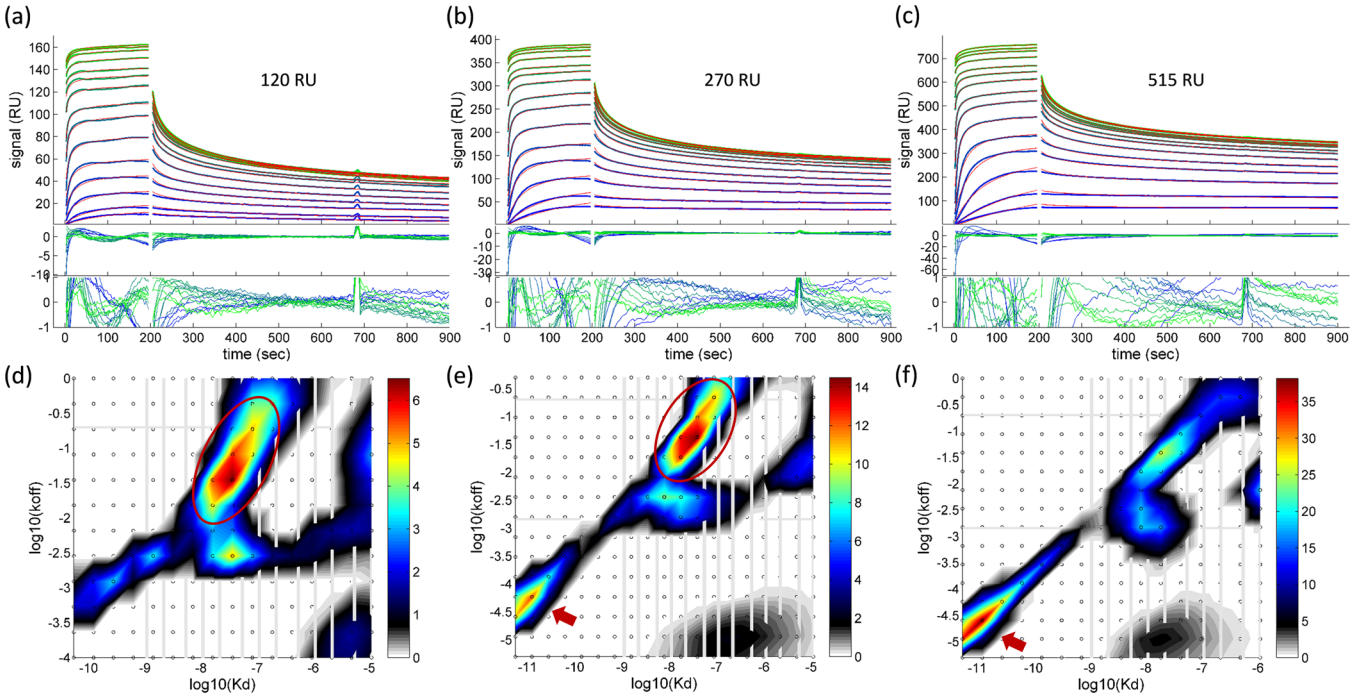


Figure 5. AuMBA−thrombin interactions by surface plasmon resonance. Experimental binding traces (green and blue lines), best-fit curves (red lines), and fitting residuals are shown for surfaces containing (a) 120, (b) 270, and (c) 515 RU of immobilized thrombin. (d−f) Calculated affinity and rate constant distributions from corresponding traces shown on top. Circled regions indicate the major peaks in the distributions. Integration of the peaks provided the binding parameters K_D, k_{on}, and k_{off}. An artifactual high-affinity peak (K_D ~pM) appears in the distributions calculated from the surfaces containing the highest immobilization densities (arrows).

Table 1. Affinity and Kinetic Parameters for AuMBA− and AuGSH−Thrombin Interactions Determined by SPR Biosensing[a]

parameter	AuMBA−thrombin	AuGSH−thrombin
K_D (M)	$(3.1 \pm 0.9) \times 10^{-8}$	$(1.9 \pm 0.7) \times 10^{-5}$
k_{on} (M^{-1} s^{-1})	$(2.0 \pm 0.4) \times 10^{6}$	$(2.8 \pm 1.5) \times 10^{4}$
k_{off} (s^{-1})	$(6.2 \pm 1.8) \times 10^{-2}$	$(5.0 \pm 2.0) \times 10^{-1}$

[a]Errors are standard deviations from five and three measurements for AuMBA and AuGSH, respectively.

relative differences in the values of k_{on} and k_{off} for AuMBA versus AuGSH binding to thrombin.

At closer look, a comparison of the affinity and rate constant distributions as a function of immobilization density for AuMBA revealed both the gradual disappearance of the major peak at 30 nM and the appearance of an artifactual high-affinity peak in the pM range (Figure 5). It can be noticed that the high-affinity peak was essentially the result of a very low rate of complex dissociation (k_{off} ~5×10^{-4} s^{-1}). One possible explanation for the appearance of the high-affinity interaction is that it was an artifact resulting from avidity effects (see Figure 4), which would reduce k_{off} significantly while causing no major changes to k_{on}. Similar binding patterns to surfaces of different ligand densities were observed for oligomeric lectins exhibiting strengthened binding with higher avidity.[48]

Another factor that could account for the high-affinity peak would be the occurrence of AuMBA rebinding during the dissociation phase; that is, AuMBA would rebind the surface before it had time to diffuse away from it. Rebinding is a manifestation of mass transport limitations when the interactions are fast ($k_{on} > 10^6$ M^{-1} s^{-1}), and therefore, this effect would be expected to occur more readily with AuMBA than with AuGSH. Mass transport limitation would be most evident from misfits of the distribution model to the initial binding and initial dissociation phase where the most rapid changes in open surface sites occur. Indeed, from inspection of the residuals for AuMBA, the largest residuals stem from these times (Figure 5). From both the consideration of possible avidity effects and mass transport limitation, the molecular parameters obtained with the low-density surface would be more trustworthy.

Binding Kinetics Probed by Fluorescence Stopped-Flow Spectroscopy. Stopped-flow spectroscopy enables reproducible and accurate measurements of the kinetics of protein−protein complexation in solution, thus avoiding surface-related avidity and mass transport effects as seen in SPR.[49] However, successful application of the stopped-flow methodology to study NP−protein interactions requires special considerations. In particular, NP−protein complexation is complicated by the fact that several proteins can bind to a single NP and establish interprotein interactions at high binding densities.[50] In addition, the orientation and/or conformation of proteins in a crowded NP surface may be different from the orientation/conformation that proteins adopt when they bind sparsely on the surface. It is therefore reasonable to expect that the association and dissociation kinetics of NP−protein binding might be influenced by the extent of the NP surface coverage with proteins. Thus, in an attempt to simplify data analysis and interpretation, the stopped-flow kinetic experiment should be ideally performed with [NP] ≫ [protein] to maintain a mostly sparsely populated NP surface.

Here, the kinetics of AuMBA−protein association was measured by stopped-flow spectroscopy under pseudo-first-order conditions. For this, solutions of FITC-labeled thrombin

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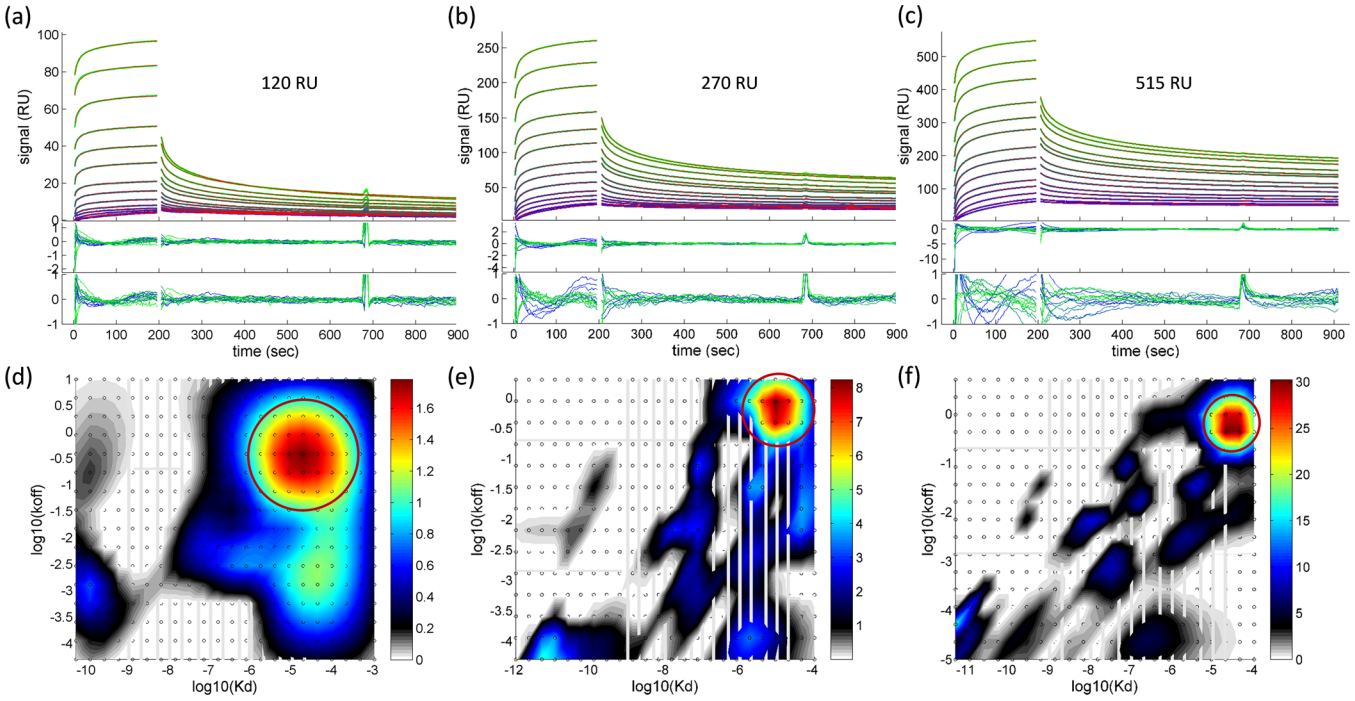


Figure 6. AuGSH−thrombin interactions by surface plasmon resonance. Experimental binding traces (green and blue lines), best-fit curves (red lines), and fitting residuals are shown for surfaces containing (a) 120, (b) 270, and (c) 515 RU of immobilized thrombin. (d−f) Calculated affinity and rate constant distributions from corresponding traces shown on top. Circled regions indicate the major peaks in the distributions. Integration of the peaks provided the binding parameters K_D, k_{on}, and k_{off}.

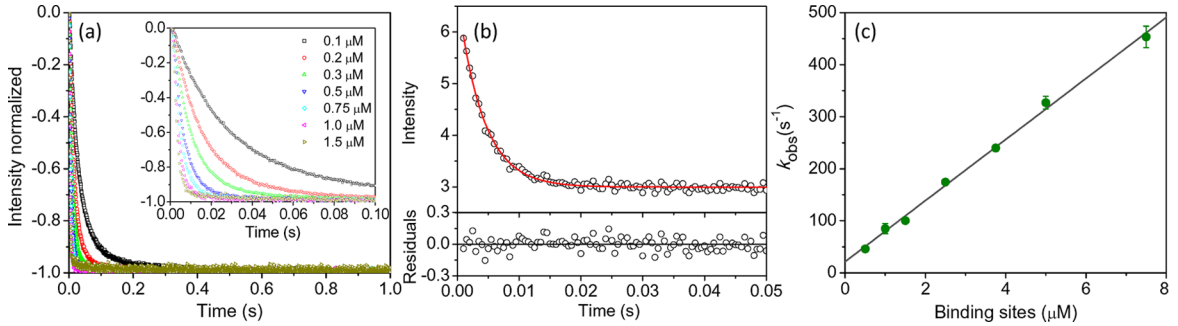


Figure 7. Stopped-flow characterization of forward binding reactions between AuMBA and thrombin. (a) Time courses of fluorescence quenching following rapid mixing of thrombin with AuMBA. Thrombin concentration was fixed to 30 nM, while nominal NP concentrations varied from 0.1 to 1.5 μM (see figure legend). (b) Single-exponential fitting to time courses yielded the observed pseudo-first-order rate constants (k_{obs}). Data illustrated for [AuMBA] = 0.75 μM. The fitted line is shown in red. (c) Plot of k_{obs} as a function of estimated total number of binding sites (assuming ~5 independent binding sites/NP). Experimental points fall on a straight line whose slope is taken as an estimate for the association rate constant (k_{on}).

Table 2. Kinetic Parameters for AuMBA− and AuGSH−Thrombin Interactions Determined by Stopped-Flow Spectroscopy

interaction with	k_{on} (M^{-1} s^{-1})	k_{off1} (s^{-1})	A_1	k_{off2} (s^{-1})	A_2	k_{off3} (s^{-1})	A_3
AuMBA	~6×10^7	0.05 ± 0.002	0.64	0.2 ± 0.03	0.19	1.8 ± 0.1	0.17
AuGSH		0.14 ± 0.01	0.37	0.64 ± 0.09	0.29	7.9 ± 0.9	0.34

were mixed with excess AuMBA, and the quenching of the FITC fluorescence signal was monitored with time (Figure 7a). A single-exponential function fitted the time traces well yielding random residuals (Figure 7b). Next, the observed pseudo-first-order rate constants (k_{obs}) obtained from the exponential fits were plotted against the concentration of AuMBA. This yielded a straight line whose slope was taken as an estimate for the association rate constant (k_{on} ~6×10^7 M^{-1} s^{-1}) (Figure 7c and Table 2). A similar analysis for AuGSH could not be performed since we were unable to obtain corresponding good quality stopped-flow data of forward reactions. It should be noted that, in calculating the k_{on} rate constant from Figure 7c, the AuMBA concentrations were converted to "number of independent binding sites" by multiplying the nominal NP concentrations by 5 (based on a previous estimate of ~6 chymotrypsin molecules bound per AuMBA at saturation[36]). On the other hand, the spatial constraints of the surface immobilization in SPR imply that a single NP can only bind to a single protein on the sensor

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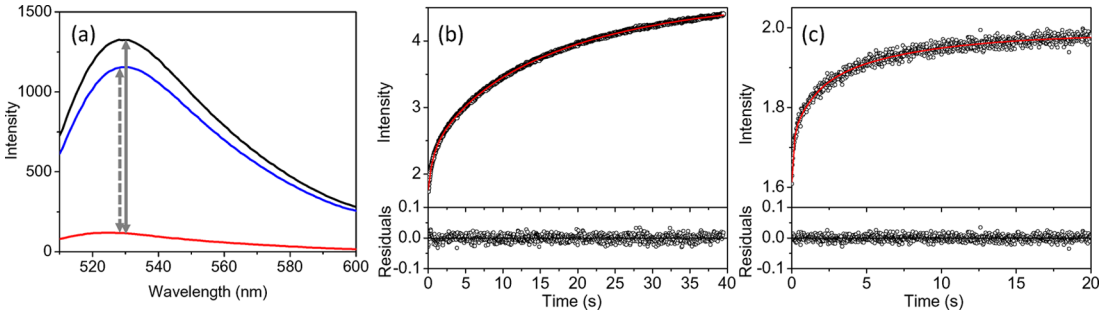


Figure 8. Characterization of NP−thrombin dissociation reactions. (a) Fluorescence emission spectra of FITC−thrombin (black trace), FITC−thrombin in the presence of AuMBA (red trace), and FITC−thrombin in the presence of AuMBA plus unlabeled thrombin in excess (blue trace). The latter spectrum was obtained 30 s after the addition of unlabeled thrombin to a solution of FITC−thrombin plus AuMBA. A comparison between the different spectra revealed that most of the initial fluorescence signal was recovered within 30 s (see dotted vs solid lines). Data was corrected for the inner-filter effect from the NPs. Similar results were obtained for AuGSH (not shown). Stopped-flow time courses for (b) AuMBA−thrombin and (c) AuGSH−thrombin reverse reactions obtained by competitive displacement of FITC-labeled thrombin from preformed complexes by unlabeled thrombin. A triple exponential function fitted the time courses with random residuals and yielded the dissociation rate constants k_{off1}, k_{off2}, and k_{off3}. Fitted lines are shown in red. FITC−thrombin, NPs, and unlabeled thrombin were used at concentrations of 50 nM, 1 μM, and 40 μM, respectively.

surface, and therefore, the nominal NP concentrations were used in the calculation of the binding constants by SPR.

The kinetics of AuMBA−thrombin and AuGSH−thrombin dissociations were evaluated by competitive displacement of FITC-labeled thrombin from preformed complexes by unlabeled thrombin. Steady-state fluorescence spectroscopy was used for a preliminary assessment of the dissociation reactions (Figure 8a). For this, emission spectra of FITC−thrombin were first obtained in the absence and presence of the NPs. Next, unlabeled thrombin was added in excess, and the final spectra were recorded after ∼30 s. A comparison between the different spectra showed that most of the initial fluorescence signal was recovered within 30 s (Figure 8a), thus implying that stopped-flow monitoring would be required to follow and quantify the dissociation reactions. Corresponding stopped-flow time courses for the AuMBA− and AuGSH−thrombin reverse reactions are therefore shown in Figure 8b,c, respectively. A triple exponential function was required to fit the time courses with random residuals. The first-order rate constants obtained from the fits equaled the dissociation rate constants k_{off1}, k_{off2}, and k_{off3} (Table 2).

The k_{on} and k_{off} values determined from the stopped-flow analysis must be strictly regarded here as apparent rate constants. This is because the calculations of both the k_{on} and k_{off} constants were affected by a lack of proportionality between the observed fluorescence signal and the extent of thrombin saturation with NPs. Specifically, FITC emission was found to be completely quenched with NP binding to one of the exosites in thrombin, whereas binding of additional particles to the remaining free exosite did not contribute additional quenching. That NP binding to a single exosite causes almost complete quenching of the fluorescence emission can be readily seen from the titrations obtained in the presence of competing HD1 or HD22 (e.g., see Figure 2a, red and blue traces).

Comparison of SPR and Stopped-Flow Results. Application of both the SPR and stopped-flow methodologies to study multisite NP−protein interactions (where multiple proteins can bind to a single NP and/or a few NPs can bind around a single protein) can be subjected to a number of experimental uncertainties and artifacts (vide supra). Nevertheless, we obtained good agreement between the kinetic parameters derived from the stopped-flow and SPR data (Tables 1 and 2). The k_{on} obtained by stopped-flow spectroscopy (∼6×10^7 M^{-1} s^{-1}) for AuMBA−thrombin was approximately 30-fold larger than that by SPR. This difference may be partly accounted for by the fact that SPR cannot reliably measure rates higher than ∼10^6 M^{-1} s^{-1}. As for the measurement of k_{off}, three dissociation rate constants were obtained from the analysis of stopped-flow reverse reactions, whereas only a single rate constant could be resolved by SPR. Nevertheless, we note that for AuMBA, k_{off1} = 0.05 s^{-1} with a phase amplitude (A_1) of 64% compares well to k_{off} = 0.06 s^{-1} obtained by SPR, whereas for AuGSH, k_{off1} = 0.14 s^{-1} and k_{off2} = 0.64 s^{-1} with a combined phase amplitude (A_1+ A_2) of 66% compare well to k_{off} = 0.5 s^{-1} from SPR. We finally note that, while the analysis of the SPR data for AuMBA−thrombin suggested a single K_D of ∼31 nM, it was not possible to assign a corresponding value (or set of values) of K_D from the stopped-flow results. This is because we could not be sure whether the interactions between AuMBA and thrombin as characterized by stopped-flow spectroscopy were defined by a single k_{on} (∼6×10^7 M^{-1} s^{-1}) or whether there were additional k_{on} values that somehow were not resolved or remained undetected.

Mechanistic Insights into Ultrasmall NP−Protein Interactions from Knowledge of k_{on}. Protein binding to ultrasmall gold NPs can be considered in analogy to a typical biomolecular interaction, as such[51,52]

$$\mathrm{P} + \mathrm{NP} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{P{:}\,NP} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} \mathrm{P{\cdot}NP} \quad (1)$$

The isolated protein and nanoparticle (P and NP) diffuse randomly in solution until they collide in the proper orientation to form an intermediate encounter complex (P:NP) with a second-order rate constant k_1. The intermediate encounter complex is mostly solvated and held together by long-range electrostatic forces. It evolves to the final bound state (P·NP) with a first-order rate constant k_2 while undergoing partial desolvation of the interface, removal of interfacial ions, and possible conformational changes.

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The observed k_{on} rate constant for a two-step reaction mechanism in which the encounter complex is assumed to be in the steady state is given by $k_{on} = k_1k_2/(k_{-1} + k_2)$.[51,52] The magnitude of the k_1 rate constant is predicted to fall close to the Smoluchowski diffusion limit for a protein interacting with spherical and uniform NPs, that is, $k_1 \sim 10^8-10^9$ M^{-1} s^{-1} at 25 °C. In comparison, the magnitude of k_1 for the association reaction between two proteins is limited by stringent orientational constraints in forming the binding interface; thus, in the absence of favorable electrostatic attraction, the maximum k_1 rate constant becomes $\sim 10^5-10^6$ M^{-1} s^{-1}.

Diffusion-limited interactions are characterized by $k_2 \gg k_{-1}$; that is, the encounter complex evolves readily to the final bound state, and the overall association rate constant reduces to $k_{on} \sim k_1$. Reaction-limited interactions, in turn, are characterized by a large transition-state energy barrier between the first encounter and final states, in which case the encounter complex tends to dissociate more readily into the unbound species rather than to proceed into the final complex. In this case, $k_{-1} \gg k_2$, and the overall association rate constant becomes $k_{on} \sim k_1k_2/k_{-1} \ll k_1$. The molecular events contributing to the energy barrier in a reaction-limited interaction may include desolvation of polar/charged residues and/or conformational changes.

The measured k_{on} rate constants for the AuMBA−thrombin and AuGSH−thrombin interactions were both lower than the predicted value of $\sim 10^8-10^9$ M^{-1} s^{-1} for a purely diffusion-limited interaction (Tables 1 and 2), therefore suggesting that both binding reactions are subjected to an energy barrier that slows down association. However, given that the association rate constant for AuGSH−thrombin complexation was at least 100-fold smaller than that for AuMBA−thrombin (Table 1), it can be concluded that a much greater energy barrier defines the transition state of the AuGSH−thrombin complex.

The nature of the proposed energy barrier for NP−protein complexation was addressed previously by computer simulations for a similar, electrostatically driven interaction system composed of AuMBA/AuGSH and the model protein CrataBL.[20] The simulations revealed a larger energy barrier for AuGSH−CrataBL complexation going from the first encounter to the final state, thus implying a smaller k_2 rate consistent with the experimental observations. The molecular nature of the barrier was attributed to the penalty required to remove interfacial ions from the more complex counterion atmosphere surrounding AuGSH. Interfacial reconfiguration of the AuGSH−CrataBL complex was also proposed to contribute to the energy barrier of its transition state.

Mechanistic Insights into Ultrasmall NP−Protein Interactions from Knowledge of k_{off}. Tables 1 and 2 show that the interactions were characterized by relatively high k_{off} rates. The corresponding residence times covered the ranges from 0.6 to 20 s for AuMBA and from 0.1 to 7 s for AuGSH, and therefore, the AuGSH−thrombin complexes are held together less tightly than AuMBA−thrombin. Overall, these relatively short residence times suggest that the binding interfaces are loosely packed and permeated by water, in analogy to the molecular nature of the interface in weak transient protein−protein complexes as established from the analysis of crystal structures.[53,54] In addition, because the NP−thrombin binding interface is predominantly polar, NP−protein complexation might be considered in analogy to the association mechanism of hydrophilic protein interfaces.[55−57] Specifically, previous computer simulations have suggested that hydrophilic protein surfaces are bridged by water molecules, thus reducing the expensive desolvation penalty that would accompany the formation of a dry interface. The bridging water mediates the formation of a hydrogen-bond network across the interface and, because of its reduced dielectric properties, contributes to enhancing the magnitude of favorable electrostatic interactions.[55] It should be noted that such a presumed water-rich interface does not exclude the possibility that direct intermolecular interactions, such as hydrogen bonds or salt bridges, may also occur between some of the NP and protein surface groups that happen to fall in closer proximity.[43,58]

The finding that the NP−thrombin complexes were characterized by three distinct dissociation processes was an interesting and unexpected outcome of our measurements. On the one hand, it is possible that each particular value of k_{off} could be assigned to a single and "specific" binding orientation of thrombin onto the NPs. Alternatively, and perhaps more likely, the interactions could produce an ensemble of distinct complexes characterized by slightly different positioning and orientations of the exosites onto the surface of the NPs. In this scenario, NP dissociation from thrombin would be characterized by multiple rate constants, but which, on average, might be resolved experimentally as only a couple of kinetically distinguishable processes as seen here. In addition, slight variations in the size of the NPs, even if by a fraction of a nanometer, could result in a different extent of stabilization of the final complexes and therefore different k_{off} values and residence times. Additional experiments will be required to differentiate among these different possibilities; for example, the use of atomically precise ultrasmall gold NPs would eliminate any possible influence of size on complex stability.

CONCLUSIONS

Ultrasmall NPs may resemble globular proteins in both size and surface chemistry and undergo reversible interactions with biomolecules. This enables one to consider ultrasmall NP−protein interactions in analogy to a typical biomolecular complexation event in which two proteins collide to form a first-encounter complex, undergo partial desolvation and/or conformational changes, and finally combine to produce the final bound complex.

Here, a new set of kinetic measurements by SPR biosensing and stopped-flow spectroscopy were implemented to further advance our mechanistic understanding of NP−protein interactions in the ultrasmall size regime. Human α-thrombin was employed as a unique and attractive model system in our investigations. This protein contains two well-defined positively charged exosite domains where negatively charged ultrasmall NPs can bind, whereas its active site can be derivatized with either FITC or biotin for optimal implementation of the fluorescence quenching and SPR methodologies.

Analysis of the SPR data revealed that the association rate constant for thrombin binding to AuGSH was ~100-fold smaller than that for AuMBA, therefore suggesting important differences in the energetics of binding along the association pathway. Presumably, the smaller k_{on} found for AuGSH is the result of a greater energy penalty required to remove the counterions surrounding AuGSH. Stopped-flow measurements revealed that the corresponding dissociation rate constants were relatively high, thus implying a weakly adhesive binding interface (overall residence times covered the range from ~0.1

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to 16 s). The reversible, transient character of ultrasmall NP−protein interactions is a key feature that distinguishes ultrasmall from large particles, and to our knowledge, this is the first study to establish the transient nature of the binding reaction quantitatively by means of highly sensitive stopped-flow characterization. In addition, the stopped-flow analysis suggested that the interactions yielded a heterogeneous population of NP−protein complexes stabilized to a different extent and therefore characterized by different residence times.

Finally, we note that knowledge of binding kinetics may also have practical implications for biomedicine, especially in the context of the open nonequilibrium in vivo system where the concentrations of NPs and proteins are in constant flux.[59,60] For example, the cellular uptake of NPs in vivo may depend not only on the concentration of NPs in the surroundings (influenced by the local mechanisms of NP delivery and clearance) but also on the characteristic timescales of NP binding with cell surface receptors (influenced by concentrations, k_{on}, and k_{off}). Therefore, engineering not only the affinity but also the kinetics of ultrasmall NP interactions could promote new ways in which to control protein function and cell behavior using nanomaterials.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) Albanese, A.; Tang, P. S.; Chan, W. C. W. The Effect of Nanoparticle Size, Shape, and Surface Chemistry on Biological Systems. *Annu. Rev. Biomed. Eng.* **2012**, *14*, 1−16.

(2) Mu, Q.; Jiang, G.; Chen, L.; Zhou, H.; Fourches, D.; Tropsha, A.; Yan, B. Chemical Basis of Interactions Between Engineered Nanoparticles and Biological Systems. *Chem. Rev.* **2014**, *114*, 7740−7781.

(3) Kotov, N. A. Inorganic Nanoparticles as Protein Mimics. *Science* **2010**, *330*, 188−189.

(4) Lira, A. L.; Ferreira, R. S.; Torquato, R. J.; Oliva, M. L. V.; Schuck, P.; Sousa, A. A. Allosteric Inhibition of α-Thrombin Enzymatic Activity with Ultrasmall Gold Nanoparticles. *Nanoscale Adv.* **2019**, *1*, 378−388.

(5) Shao, Q.; Hall, C. K. Allosteric Effects of Gold Nanoparticles on Human Serum Albumin. *Nanoscale* **2017**, *9*, 380−390.

(6) Kopp, M.; Kollenda, S.; Epple, M. Nanoparticle−Protein Interactions: Therapeutic Approaches and Supramolecular Chemistry. *Acc. Chem. Res.* **2017**, *50*, 1383−1390.

(7) Hassan, S. A. Strong Dependence of the Nano-Bio Interactions on Core Morphology and Layer Composition of Ultrasmall Nanostructures. *J. Chem. Phys.* **2019**, *151*, 105102.

(8) An, D.; Su, J.; Weber, J. K.; Gao, X.; Zhou, R.; Li, J. A Peptide-Coated Gold Nanocluster Exhibits Unique Behavior in Protein Activity Inhibition. *J. Am. Chem. Soc.* **2015**, *137*, 8412−8418.

(9) Kim, B. H.; Hackett, M. J.; Park, J.; Hyeon, T. Synthesis, Characterization, and Application of Ultrasmall Nanoparticles. *Chem. Mat.* **2013**, *26*, 59−71.

(10) Jiang, Y.; Huo, S.; Mizuhara, T.; Das, R.; Lee, Y.-W.; Hou, S.; Moyano, D. F.; Duncan, B.; Liang, X.-J.; Rotello, V. M. The Interplay of Size and Surface Functionality on the Cellular Uptake of Sub-10 nm Gold Nanoparticles. *ACS Nano* **2015**, *9*, 9986−9993.

(11) Porret, E.; Sancey, L.; Martín-Serrano, A.; Montañez, M. I.; Seeman, R.; Yahia-Ammar, A.; Okuno, H.; Gomez, F.; Ariza, A.; Hildebrandt, N.; Fleury, J.-B.; Coll, J.-L.; le Guével, X. Hydrophobicity of Gold Nanoclusters Influences Their Interactions with Biological Barriers. *Chem. Mat.* **2017**, *29*, 7497−7506.

(12) Treuel, L.; Nienhaus, G. U. Toward a Molecular Understanding of Nanoparticle−Protein Interactions. *Biophy. Rev.* **2012**, *4*, 137−147.

(13) Casals, E.; Pfaller, T.; Duschl, A.; Oostingh, G. J.; Puntes, V. Time Evolution of the Nanoparticle Protein Corona. *ACS Nano* **2010**, *4*, 3623−3632.

(14) Lundqvist, M.; Sethson, I.; Jonsson, B.-H. Protein Adsorption onto Silica Nanoparticles: Conformational Changes Depend on the Particles' Curvature and the Protein Stability. *Langmuir* **2004**, *20*, 10639−10647.

(15) Wang, H.; Lin, Y.; Nienhaus, K.; Nienhaus, G. U. The Protein Corona on Nanoparticles as Viewed from a Nanoparticle-Sizing Perspective. *Wiley Intern. Rev. Nanomed. Nanobiotech.* **2018**, *10*, No. e1500.

(16) Wang, A.; Vangala, K.; Vo, T.; Zhang, D.; Fitzkee, N. C. A Three-Step Model for Protein−Gold Nanoparticle Adsorption. *J. Phys. Chem. C* **2014**, *118*, 8134−8142.

(17) Piella, J.; Bastús, N. G.; Puntes, V. Size-Dependent Protein−Nanoparticle Interactions in Citrate-Stabilized Gold Nanoparticles: The Emergence of the Protein Corona. *Bioconj. Chem.* **2016**, *28*, 88−97.

(18) Dennison, J. M.; Zupancic, J. M.; Lin, W.; Dwyer, J. H.; Murphy, C. J. Protein Adsorption to Charged Gold Nanospheres as a Function of Protein Deformability. *Langmuir* **2017**, *33*, 7751−7761.

(19) Boselli, L.; Polo, E.; Castagnola, V.; Dawson, K. A. Regimes of Biomolecular Ultrasmall Nanoparticle Interactions. *Angew. Chem. Int. Ed.* **2017**, *56*, 4215−4218.

(20) Lira, A. L.; Ferreira, R. S.; Torquato, R. J. S.; Zhao, H.; Oliva, M. L. V.; Hassan, S. A.; Schuck, P.; Sousa, A. A. Binding Kinetics of Ultrasmall Gold Nanoparticles with Proteins. *Nanoscale* **2018**, *10*, 3235−3244.

(21) Bekdemir, A.; Stellacci, F. A Centrifugation-Based Physicochemical Characterization Method for the Interaction Between Proteins and Nanoparticles. *Nat. Commun.* **2016**, *7*, 13121.

(22) Hühn, J.; Fedeli, C.; Zhang, Q.; Masood, A.; del Pino, P.; Khashab, N. M.; Papini, E.; Parak, W. J. Dissociation Coefficients of Protein Adsorption to Nanoparticles as Quantitative Metrics for Description of the Protein Corona: A Comparison of Experimental Techniques and Methodological Relevance. *Int. J. Biochem. Cell Biol.* **2016**, *75*, 148−161.

(23) Boulos, S. P.; Davis, T. A.; Yang, J. A.; Lohse, S. E.; Alkilany, A. M.; Holland, L. A.; Murphy, C. J. Nanoparticle−Protein Interactions: A Thermodynamic and Kinetic Study of the Adsorption of Bovine Serum Albumin to Gold Nanoparticle Surfaces. *Langmuir* **2013**, *29*, 14984−14996.

(24) De, M.; You, C.-C.; Srivastava, S.; Rotello, V. M. Biomimetic Interactions of Proteins with Functionalized Nanoparticles: A Thermodynamic Study. *J. Am. Chem. Soc.* **2007**, *129*, 10747−10753.

(25) Engel, M. F. M.; van Mierlo, C. P. M.; Visser, A. J. W. G. Kinetic and Structural Characterization of Adsorption-Induced Unfolding of Bovine α-Lactalbumin. *J. Biol. Chem.* **2002**, *277*, 10922−10930.

(26) Pan, H.; Qin, M.; Meng, W.; Cao, Y.; Wang, W. How Do Proteins Unfold Upon Adsorption on Nanoparticle Surfaces? *Langmuir* **2012**, *28*, 12779−12787.

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DOI: 10.1021/acs.jpcc.9b08308
J. Phys. Chem. C 2019, 123, 28450−28459

(27) Prado-Gotor, R.; Grueso, E. A Kinetic Study of the Interaction of DNA with Gold Nanoparticles: Mechanistic Aspects of the Interaction. *Phy. Chem. Chem. Phys.* **2011**, *13*, 1479−1489.
(28) Patra, A.; Ding, T.; Engudar, G.; Wang, Y.; Dykas, M. M.; Liedberg, B.; Kah, J. C. Y.; Venkatesan, T.; Drum, C. L. Component-Specific Analysis of Plasma Protein Corona Formation on Gold Nanoparticles Using Multiplexed Surface Plasmon Resonance. *Small* **2016**, *12*, 1174−1182.
(29) Huntington, J. A. Thrombin Plasticity. *Biochim. Biophy. Acta Prot. Proteom.* **2012**, *1824*, 246−252.
(30) Bock, P. E. Active Site Selective Labeling of Serine Proteases with Spectroscopic Probes Using Thioester Peptide Chloromethyl Ketones: Demonstration of Thrombin Labeling Using N. Alpha. [(Acetylthio) Acetyl]-D-Phe-Pro-Arg-Ch2cl. *Biochemistry* **2002**, *27*, 6633−6639.
(31) Verespy, S., III; Mehta, A. Y.; Afosah, D.; Al-Horani, R. A.; Desai, U. R. Allosteric Partial Inhibition of Monomeric Proteases. Sulfated Coumarins Induce Regulation, Not Just Inhibition, of Thrombin. *Sci. Rep.* **2016**, *6*, 24043.
(32) Knittel, L. L.; Schuck, P.; Ackerson, C. J.; Sousa, A. A. Zwitterionic Glutathione Monoethyl Ester as a New Capping Ligand for Ultrasmall Gold Nanoparticles. *RSC Adv.* **2016**, *6*, 46350−46355.
(33) Sousa, A. A.; Hassan, S. A.; Knittel, L. L.; Balbo, A.; Aronova, M. A.; Brown, P. H.; Schuck, P.; Leapman, R. D. Biointeractions of Ultrasmall Glutathione-Coated Gold Nanoparticles: Effect of Small Size Variations. *Nanoscale* **2016**, *8*, 6577−6588.
(34) Ackerson, C. J.; Powell, R. D.; Hainfeld, J. F. In *Methods in Enzymology*; Jensen, G. J., Eds.; Academic Press: New York, 2010; Vol. *481*, pp 195−230.
(35) Heinecke, C. L.; Ackerson, C. J. In *Nanoimaging. Methods in Molecular Biology (Methods and Protocols)*; Sousa, A., Kruhlak, M., Eds.; Humana Press: 2013; Vol. *950*, pp 293−311.
(36) Sousa, A. A. A Note on the Use of Steady−State Fluorescence Quenching to Quantify Nanoparticle−Protein Interactions. *J. Fluoresc.* **2015**, *25*, 1567−1575.
(37) Sousa, A. A. In *Reviews in Fluorescence 2017*; Geddes, C., Ed.; Springer: 2018; pp 53−73.
(38) Schuck, P.; Zhao, H. In *Surface Plasmon Resonance*; Mol, N. J., Fischer, M. J. E., Eds.; Humana Press: 2010; Vol. *627*, pp 15−54.
(39) Svitel, J.; Balbo, A.; Mariuzza, R. A.; Gonzales, N. R.; Schuck, P. Combined Affinity and Rate Constant Distributions of Ligand Populations from Experimental Surface Binding Kinetics and Equilibria. *Biophys. J.* **2003**, *84*, 4062−4077.
(40) Svitel, J.; Boukari, H.; Van Ryk, D.; Willson, R. C.; Schuck, P. Probing the Functional Heterogeneity of Surface Binding Sites by Analysis of Experimental Binding Traces and the Effect of Mass Transport Limitation. *Biophys. J.* **2007**, *92*, 1742−1758.
(41) Stubbs, M. T.; Bode, W. The Clot Thickens: Clues Provided by Thrombin Structure. *Trends Biochem. Sci.* **1995**, *20*, 23−28.
(42) Jurrus, E.; Engel, D.; Star, K.; Monson, K.; Brandi, J.; Felberg, L. E.; Brookes, D. H.; Wilson, L.; Chen, J.; Liles, K.; Chun, M.; Li, P.; Gohara, D. W.; Dolinsky, T.; Konecny, R.; Koes, D. R.; Nielsen, J. E.; Head-Gordon, T.; Geng, W.; Krasny, R.; Wei, G.-W.; Holst, M. J.; McCammon, J. A.; Baker, N. A. Improvements to the APBS Biomolecular Solvation Software Suite. *Prot. Sci.* **2018**, *27*, 112−128.
(43) Lin, W.; Insley, T.; Tuttle, M. D.; Zhu, L.; Berthold, D. A.; Král, P.; Rienstra, C. M.; Murphy, C. J. Control of Protein Orientation on Gold Nanoparticles. *J. Phys. Chem. C* **2015**, *119*, 21035−21043.
(44) Hassan, S. A. Computational Study of the Forces Driving Aggregation of Ultrasmall Nanoparticles in Biological Fluids. *ACS Nano* **2017**, *11*, 4145−4154.
(45) Bayraktar, H.; You, C.-C.; Rotello, V. M.; Knapp, M. J. Facial Control of Nanoparticle Binding to Cytochrome c. *J. Am. Chem. Soc.* **2007**, *129*, 2732−2733.
(46) Breitsprecher, D.; Schlinck, N.; Witte, D.; Duhr, S.; Baaske, P.; Schubert, T. In *Nucleic Acid Aptamers. Methods in Molecular Biology*; Mayer, G., Eds.;. Humana Press: New York 2016; Vol. *1380*, pp 99−111.
(47) Pica, A.; Russo Krauss, I.; Parente, V.; Tateishi-Karimata, H.; Nagatoishi, S.; Tsumoto, K.; Sugimoto, N.; Sica, F. Through-Bond Effects in the Ternary Complexes of Thrombin Sandwiched by Two DNA Aptamers. *Nucleic Acids Res.* **2017**, *45*, 461−469.
(48) Gjelstrup, L. C.; Kaspersen, J. D.; Behrens, M. A.; Pedersen, J. S.; Thiel, S.; Kingshott, P.; Oliveira, C. L. P.; Thielens, N. M.; Vorup-Jensen, T. The Role of Nanometer-Scaled Ligand Patterns in Polyvalent Binding by Large Mannan-Binding Lectin Oligomers. *J. Immunol.* **2012**, *188*, 1292−1306.
(49) Zheng, X.; Bi, C.; Li, Z.; Podariu, M.; Hage, D. S. Analytical Methods for Kinetic Studies of Biological Interactions: A Review. *J. Pharm. Biomed. Anal.* **2015**, *113*, 163−180.
(50) Bhirde, A. A.; Hassan, S. A.; Harr, E.; Chen, X. Role of Albumin in the Formation and Stabilization of Nanoparticle Aggregates in Serum Studied by Continuous Photon Correlation Spectroscopy and Multiscale Computer Simulations. *J. Phys. Chem. C* **2014**, *118*, 16199−16208.
(51) Schreiber, G.; Haran, G.; Zhou, H.-X. Fundamental Aspects of Protein− Protein Association Kinetics. *Chem. Rev.* **2009**, *109*, 839−860.
(52) Zhou, H.-X.; Bates, P. A. Modeling Protein Association Mechanisms and Kinetics. *Curr. Op. Struc. Biol.* **2013**, *23*, 887−893.
(53) Perkins, J. R.; Diboun, I.; Dessailly, B. H.; Lees, J. G.; Orengo, C. Transient Protein-Protein Interactions: Structural, Functional, and Network Properties. *Structure* **2010**, *18*, 1233−1243.
(54) Rodier, F.; Bahadur, R. P.; Chakrabarti, P.; Janin, J. Hydration of Protein−Protein Interfaces. *Prot. Struc. Func. Bioinf.* **2005**, *60*, 36−45.
(55) Ahmad, M.; Gu, W.; Geyer, T.; Helms, V. Adhesive Water Networks Facilitate Binding of Protein Interfaces. *Nat. Commun.* **2011**, *2*, 261.
(56) Ulucan, O.; Helms, V. How Hydrophilic Proteins Form Nonspecific Complexes. *J. Phys. Chem. B* **2015**, *119*, 10524−10530.
(57) Ulucan, O.; Jaitly, T.; Helms, V. Energetics of Hydrophilic Protein−Protein Association and the Role of Water. *J. Chem. Theory Comput.* **2014**, *10*, 3512−3524.
(58) Tollefson, E. J.; Allen, C. R.; Chong, G.; Zhang, X.; Rozanov, N. D.; Bautista, A.; Cerda, J. J.; Pedersen, J. A.; Murphy, C. J.; Carlson, E. E.; Hernandez, R. Preferential Binding of Cytochrome c to Anionic Ligand-Coated Gold Nanoparticles: A Complementary Computational and Experimental Approach. *ACS Nano* **2019**, *13*, 6856−6866.
(59) Copeland, R. A. The Drug−Target Residence Time Model: A 10-Year Retrospective. *Nat. Rev. Drug Discov.* **2016**, *15*, 87.
(60) Sousa, A. A. Impact of Soft Protein Interactions on the Excretion, Extent of Receptor Occupancy and Tumor Accumulation of Ultrasmall Metal Nanoparticles: A Compartmental Model Simulation. *RSC Adv.* **2019**, *9*, 26927−26941.

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